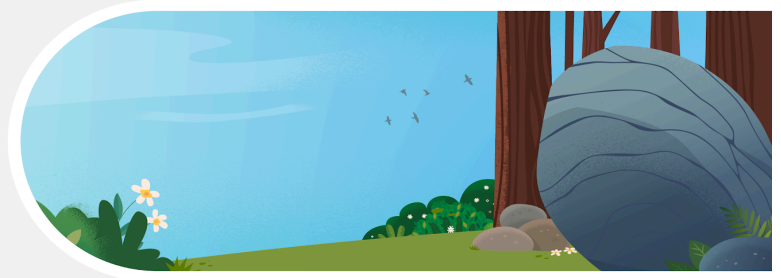


Bedrock Domain Scenario



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Data Architecture and Management

Instructions

Presentation Time : 20 minutes

Q & A : 25 minutes

Immediate Feedback : 15 minutes

For the hypothetical scenario described below, you will need to document and present your proposed solution design followed by a Q&A session with a judge. As part of the solution you should:

- Describe an appropriate solution that meets the business requirements
- Deliver all of the required deliverables as part of that presentation
- Address trade-offs and considerations for your solution options

Your presentation should be tailored for your target audience <Enter Role>. In cases where requirements are not explicitly stated for the scenario, you should use your best judgment and make appropriate assumptions based on the information provided in the Bedrock Overview (Domain Zero). You should indicate any assumptions that were made when designing the proposed solution. You will not have an opportunity to ask clarifying questions related to the hypothetical scenario.

You will be evaluated on your ability to assess the scenario requirements, design a solution, communicate the proposed architecture, and justify the design decisions.

The executive will evaluate your solution in the following areas:

Objective	Weighting
Design and Optimization	15%
Data Movement	10%
Data Quality	13%
Data Governance & Stewardship	8%
Master Data Management	10%
Reference Data Management	10%
Data Archiving	9%
Data Migration	15%

Data Solution Architecture Definition	10%
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Expected Deliverables

Please use the following template to guide your presentation. You may add slides, but only if required to further explain your solution. The solution should not have more than 5 slides. Functioning demo of the core components of your solution.

The executive who you are presenting to is very busy. Please ensure that you only schedule time during the windows s/he has made available.

Problem Overview

The transportation division of Bedrock has an Automobile business, Bedrock Auto (BA), which sells Used cars procured through Fleet and other Auctions to Dealers who then sell the vehicles to end consumers. The business operates primarily in the 50 US states. This business segment is expected to grow by 20% year over year for the next 5 years. Currently, there are 10K Dealerships nationwide affiliated with BA and about 2M consumers who own vehicles purchased through their dealerships.

BA is an existing Salesforce Sales Cloud user and is in the process of implementing their next generation customer service capabilities using Salesforce Service Cloud.

The current systems in their architecture are as follows:

1. **Legacy Service CRM** - This is a highly customized legacy Case Management application that will be replaced by Service Cloud. BA generates over 500K cases per year from consumers and dealers. This system currently houses the last 2-years of Cases and supports only file-based integration. The challenge lies in efficiently extracting, transforming, and loading this historical data while preserving its integrity and establishing ongoing synchronization for a period during the transition.
2. **ERP** - The ERP system is their master vehicle repository and currently stores over 20M vehicles. It also has a repository of Dealers, but they have significant issues with duplicates and inconsistent data in this area. ERP keeps track of warranties purchased and supports Invoices for parts orders. This system provides web services for integration, but the sheer volume and complexity of data transformations required for true data harmonization are proving challenging with point-to-point integrations alone.
3. **Parts System** - This is a home-grown Java web application that stores their parts catalog and lets dealers order parts and process orders. The system stores dealer profile information as well. There are over 500K separate parts across multiple year, make, model of vehicles. There can be over 500k orders per year, each generating invoices in the ERP system once the order is placed. Each order has an average of 3 line items. This system provides a UI to search and browse the parts catalog and place orders. It can support some modifications if needed, as it is maintained by in-house Java

developers. BA wants dealers to use Salesforce as their front-end to access these functions going forward. The system provides web services to place orders and get order status. However, ensuring data consistency and timeliness across this system, Salesforce, and the ERP requires a robust data integration strategy.

4. **ESB** - BA has purchased a modern ESB platform to support their integration needs but have not fully leveraged this technology to justify the investment. The CIO wants the ESB to be highly leveraged for maintainable integrations and quicker time to market, particularly for real-time operational data flows. However, for large-scale data migrations, complex transformations, and batch processing, additional tooling may be necessary to complement the ESB's capabilities.
5. **ETL** - BA previously considered building and maintaining all ETL functionality in Java, but given the escalating data volumes, diversity of sources, and increasing complexity of data governance and quality requirements, they are now re-evaluating the strategic necessity of an enterprise-grade ETL tool. They need to understand how such a tool can streamline complex data extraction, transformation, loading, and orchestration processes beyond what custom code or the ESB can efficiently handle.
6. **EDW** - BA has a modern Oracle data warehouse that they use to archive non-operational data for analytical reporting and trending purposes. While valuable for historical analysis, the EDW's batch nature and schema rigidity limit its ability to provide real-time, unified customer insights for operational use cases directly within Salesforce or for cross-cloud analytics.

The "actors" the system should support are:

1. **Service Agents** - Handle cases through phone, email, web, chat, and social channels. These agents are regionally based and support inquiries and cases for Dealers and Consumers in their region (50 broad regions, 10 agents per region). Service agents also create articles and associate these to cases. They require a consolidated view of customer interactions and historical data, regardless of the data source.
2. **Special Agents** - Fluid groups of 3 specialists handling complex cases, with teams in each region. Service Agents can be moved into a Specialist team and vice-versa depending on the need.
3. **Regional Service Managers** - Manage service agents for their region, needing access to Cases within their regions and managing Dealers in their region. They require

real-time visibility into agent performance and regional service trends, drawing data from various systems.

4. **Service VP** - Needs cross-region Case reporting and high-level operational insights.
5. **Service Operations** - This team focuses on analyzing demand and proposing promotions to increase utilization of BA's own service centers. They need agile access to diverse operational data, including service records, vehicle data, and customer interactions, for proactive analysis and decision-making, which is currently hindered by siloed systems and weekly data updates.
6. **Dealers** - 10K dealerships affiliated with BA, each with up to 5 users. Dealers should be able to look at cases assigned to them and collaborate with BA staff to solve these cases. Dealers can also submit cases assigned based on their region. They can also look at their invoices and submit parts orders. Dealer Users are dealership service consultants (DSC) who report to Dealership Service Managers (DSM). DSMs need access to all records owned by the DSC and the ability to add new DSC users and manage their contact information. A persistent challenge is providing dealers with a consistent, accurate, and up-to-date view of their data, which is fragmented across multiple internal systems.
7. **Consumers** - Over 2M consumers who need the ability to submit cases for their vehicles, log product issues, and order parts. Consumers can also search and view Articles related to Vehicle Models as well as Solutions to some Do-It-Yourself fixes for common vehicle problems. To deliver a truly personalized customer experience, BA recognizes the need to unify customer data from all interaction points – website, mobile app, service calls, and vehicle data – into a single, actionable customer profile.

System and Process Requirements:

Service Processes:

Their legacy CRM application is out of support and unable to meet the new capabilities required by the customer service agents and managers. Due to compliance and regulatory reasons, one of the key requirements is to be able to access past 7 years of historical Case data from the Salesforce platform. This includes Cases, comments, and related entities including case ownership. While 2 years of operational cases can reside in Salesforce, up to 7 years of data must be seamlessly accessible for viewing, requiring an effective strategy for long-term data

retention and access. The legacy CRM application will be decommissioned 3-months after the data and application both are migrated to Salesforce. During these 3 months, changes to current cases in both systems need to be synchronized, necessitating a robust bi-directional data flow. Additionally, a review of all cases suggested that there are several closed cases taking up valuable storage space on Salesforce, highlighting a clear need for an archiving strategy that balances accessibility with operational efficiency.

Following this migration, Bedrock Enterprises will continue to use their ERP system for Invoicing and Payments. In the past, legacy CRM users had to log in to the ERP system separately to see the latest invoices and payment status, leading to average call time of 3-5 minutes, where the majority of the time was spent looking for the correct ERP account and finding the latest status. In the new system, BA wants Service reps to be able to see Customers' Invoice and payment information without having to go to the ERP system, requiring a real-time or near real-time consolidation of critical financial data with customer service interactions.

During discovery meetings, Service Managers expressed frustration on the quality of Dealership information in the application as the root cause of why their calls take a lot longer than they would like. Even when they find the right record and discover data quality issues, they do not currently have any ability to correct the data and synchronize the updates back to their ERP. BA wants to fix these data issues related to duplicates and standardization as part of this migration effort, demanding sophisticated data quality, matching, and reconciliation capabilities across systems. Currently, system owners serve as data stewards for their domains. BA has been looking into establishing a central Data Governance body, but that initiative is not gaining a lot of traction. BA is evaluating alternative strategies to institute data governance to ensure consistent standards and rules are established, and appropriate controls are in place to ensure adherence to these standards and rules. Additionally, an evaluation is underway to select a 3rd party data provider to improve data accuracy and completeness, necessitating a robust mechanism for ingesting, mapping, and harmonizing this external data with existing internal data sources.

You've also discovered that Bedrock's Service Center network regularly updates the vehicle service records obtained from affiliate dealerships into the ERP system where these records are

associated with the vehicle. BA would like to leverage these during case-handling to incentivize owners or lessees to purchase additional maintenance programs or extend their warranties for their vehicles. This requires the creation of a unified customer profile that brings together vehicle service history, warranty information, and customer interaction data from disparate sources to enable proactive engagement and personalized offers.

Dealer Processes :

Dealerships need the ability to log in and look up vehicles by VIN and customer to find current warranty information from the system. They also need the ability to look up Parts and submit Part orders through the new system. The part order needs to be processed by the Parts system, which can provide Order status, delivery date, etc. In addition, Dealer users need the ability to submit Inquiries about Orders to BA. Dealer Managers should be able to get a report of vehicles serviced by them as well as Case reports on cases their dealership has handled.

Dealers have expressed dissatisfaction with delays in receiving parts orders from Bedrock. There have been many occasions when their orders were shipped to incorrect addresses, causing delays. Dealer information exists in multiple systems, and ensuring the most complete and accurate information is used in transactions has been a pervasive challenge for BA, highlighting the critical need for a centralized, trusted master data record for all dealers.

Customer Processes :

Customers need to be able to log in to the system and get information about the vehicles they own as well as their vehicle maintenance records. They also need to be able to submit Cases around issues and concerns with their vehicle. They can also submit complaints about Dealerships. The Dealership complaints should be handled by a special group of service agents by region and are not visible to anyone else in the system. Customers can also search and view Articles related to Vehicle Models as well as Solutions to some Do-It-Yourself fixes for common vehicle problems. To deliver a truly personalized customer experience, BA recognizes the need to unify customer data from all interaction points – website, mobile app, service calls, and vehicle data – into a single, actionable customer profile.

Account Executive Needs:

Account Executives have expressed concerns as well. They are unable to see the total amount of business generated by recurring billings, as this data is locked behind spreadsheets finance produces based on ERP data marts. They also raised concerns that service inefficiencies caused some of the larger business customers' concerns. Account Executives would like to see a holistic 360-degree view of their large business customers to serve better to their needs, which necessitates aggregating and activating data from Sales Cloud, Service Cloud, ERP, and other external systems.

Account execs would also like to understand the financial impact of post-process cleanup. For example, delays in order shipments to dealers as orders were shipped to inaccurate shipping addresses. What does it cost BA to correct this error post-process? What Data Quality metrics should BA be tracking, and how can these metrics be linked to enterprise financial measures? This requires advanced data observability and the ability to correlate data quality issues with business impact.

Operations Needs:

The Operations team expressed frustration that relying on structured reports on top of the data warehouse has not met their needs as case information is updated and summarized only once a week. They require near real-time operational dashboards and alerts to identify anomalies across product regions or dealers, expose this information to call center managers who may need to make scheduling adjustments, and include dealer metrics front and center. This demands a data platform capable of ingesting high-volume, streaming data and providing immediate, actionable insights for operational decision-making.

Other Needs:

BA cannot afford to lose any data due to any unforeseen events. So, recommending a robust backup strategy to meet their compliance and business continuity requirements, including considerations for point-in-time recovery and disaster recovery, is critical. BA is looking forward

to hearing your approach to designing a backup & recovery strategy including how you will approach extracting data from Salesforce without impacting other operations and best practices from a data management perspective.

BA requires 5 years of historical data to be available on the new platform. Any customer-related data older than 5 years and up to 7 years must be available on demand for compliance and reporting, but not necessarily stored directly within the operational Salesforce instance. This mandates a tiered data management strategy that includes both near-line and archival solutions.

BA is also evaluating their master data management strategy. They are looking at build vs. buy options and also considering cloud-based options. They'd like to get a better handle on contextual data to ensure consistency in their transactions. Recommend a strategy and approach towards driving BA's customer data harmonization across all systems to achieve a single source of truth and a unified customer identity. This includes addressing the complexities of identity resolution and data enrichment from various touchpoints.

BA shared, in confidence, that their business eyes expansion into European markets so there's an emerging need to potentially manage multiple CRM environments whilst maintaining a consolidated, cross-regional view of the entire book of business. Whilst the current focus is on the enhancements of its customer engagement strategy and operational intelligence in the US, BA would like to hear how you would approach a global solution that should support scalable customer data unification, enable advanced segmentation and personalization, and empower their Service Agents with AI-driven recommendations.

Estimated Datasets:

- 20M VINS
- 100M Service Records
- 2.5M Cases
- 1M Orders/year
- 3 Order line items/year
- 15K Dealers, both company-owned and franchisees

- 75K Dealers' employees
- 2M Consumer accounts
- 100K Attachments
- 50K Knowledge Articles

Solution Description

You are engaged to provide the customer with a holistic approach of how their Customer Data can be migrated and structured through the Salesforce platform, such that historical data quality concerns can be addressed in a sustained and systematic way. Leverage leading best practices for data management, data governance, integration architectures, and migration in order to address customer's needs, recognizing they produce significant amounts of data through their processes.

The Solution should address the following:

1. Based on the Actor roles and their specific needs, list the Licenses you will use to support these roles. Also, provide a high-level Role hierarchy. [Solution Architecture]
2. Estimate data volumes for key entities over a 2-year horizon.
3. A recommended Data model showing the entities and relationships (M-D, lookups, External) needed. Indicate ownership, OWD for all key entities along with an indicator for entities with LDV considerations. Also, indicate any off-platform entities. [Data Model, Design and Optimization]
4. System Landscape including the systems as well as the platforms you will be using in the Solution. [Solution Architecture]
5. Data Migration Processes needed for successful one-time as well as ongoing data synchronization where applicable. [Data Migration, Data Movement]
6. Solution for LDV scenarios related to querying, searching, and accessing data within the operational Service Cloud instance, considering both within-platform and external data access strategies. [Solution Architecture, Design and Optimization]